

Full-Stack Application Development

Welcome to the course

Week 1, Day 1 — Web Foundations



Instructor

Dilovan Matini



Today's Focus

Web Foundations

Course Overview

Your journey into full-stack development starts here



10-Week Program

Comprehensive full-stack development curriculum designed for beginners



Three Core Areas

Backend

Web

Mobile



Hands-On Learning

Project-based approach with practical exercises every day



AI-Powered Development

Using Cursor IDE + AI Support

- ✓ Intelligent code completion
- ✓ Real-time error detection
- ✓ AI-assisted learning



Designed for Beginners

No prior coding experience required. We emphasize **practical understanding** over theory.

What Is Software?

Understanding the foundation of modern technology



Software = Instructions

Software is a set of **instructions** that tell computers what to do. It's the bridge between human ideas and machine execution.

📱 Everyday Examples You Use



Websites

News, shopping, social



Mobile Apps

Messaging, banking, games



Games

Entertainment & learning



Tools

Productivity & creativity



Our Focus



Web Applications

Browser-based software



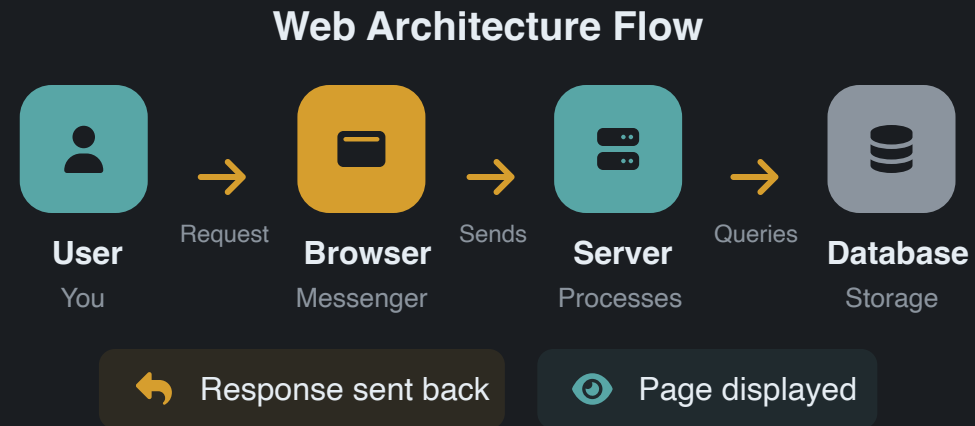
Mobile Applications

Smartphone & tablet apps

“ “Software is eating the world.” — Marc Andreessen

How the Web Works

The fundamental flow of web interactions



1 User Action

User types a URL or clicks a link in the browser

2 Browser Request

Browser sends HTTP request to the server

3 Server Response

Server processes request and sends back data

4 Page Display

Browser renders the received content for user

| Browser vs Server

Understanding the two sides of web development



Browser

The Frontend



Shows Web Pages

Renders HTML, CSS, and displays content to users



Runs Basic Code

Executes JavaScript for interactivity



Handles User Input

Captures clicks, typing, and interactions

Think of it as: The storefront that customers see and interact with



Server

The Backend



Stores Logic

Processes business rules and calculations



Manages Data

Stores and retrieves information from databases



Handles Security

Authentication, authorization, and protection

Think of it as: The stock room and management system behind the scenes



In this course, you will build **BOTH!** Frontend + Backend = Full-Stack Developer



What Is HTML?

The foundation structure of every web page



HyperText Markup Language

HTML is the **standard markup language** for documents designed to be displayed in a web browser.

🧩 HTML = Structure of a Web Page

Think of HTML as the **skeleton** of a website. It defines:

- ✓ What content appears
- ✓ Hierarchy of elements
- ✓ How it's organized
- ✓ Semantic meaning

🔖 Essential HTML Tags

<html>

Root element of the page

<head>

Metadata and page info

<body>

Visible content container

<h1>

Heading levels

<p>

Paragraph text



Analogy: Tags are like **labels** that tell the browser what each piece of content is.

Web Page Structure

The basic HTML document structure

index.html

```
<html>
  <head></head>
  <body>
    <h1>Title</h1>
    <p>Paragraph</p>
  </body>
</html>
```

Key Points

1. Every HTML file starts with
2. All content goes inside tags
3. contains metadata (not visible)
4. contains visible content

Visual Structure

— Root Container

— Metadata

- Page title
- Links to CSS/JS

— Visible Content

- Headings
- Paragraphs
- Images, links, etc.

Live Demo

We'll create this structure together in **Cursor IDE** and see it render in the browser!

What Is CSS?

Making web pages beautiful and styled



Cascading Style Sheets

CSS is a **stylesheet language** used to describe the presentation of a document written in HTML.



What CSS Controls



Colors

Text & backgrounds



Layout

Position & structure



Spacing

Margins & padding

Analogy: If HTML is the **skeleton**, CSS is the **skin, hair, and clothing** that makes it look good!

</> CSS Syntax

```
selector {  
  property : value ;  
}
```

Selector: Which element to style

Property: What to change

Value: The new setting

★ Why CSS Matters

- ✓ Separates content from design
- ✓ Reusable across pages
- ✓ Consistent branding
- ✓ Responsive design

CSS Example

See how CSS transforms your HTML

style.css

```
h1 {  
  color : blue ;  
}  
  
/* Changes heading color */  
  
p {  
  font-size : 18px ;  
}  
  
/* Sets paragraph size */
```

Linking CSS to HTML

```
<rel= "stylesheet" href= "style.css">
```

Before & After

WITHOUT CSS (Plain HTML):

Title

This is a paragraph.



WITH CSS (Styled):

Title

This is a paragraph.

Try It Yourself

Change the color to **red** or **green** . Modify the font-size to **24px** . See the changes instantly!

What Is JavaScript?

Adding interactivity and behavior to web pages



JavaScript (JS)

JavaScript is a **programming language** that enables interactive web pages and is an essential part of web applications.



What JavaScript Does



Responds to User Actions

Clicks, typing, scrolling, hovering



Updates Content Dynamically

Change text, images without reloading



Performs Calculations

Math, logic, data processing



Communicates with Servers

Fetch data, send information



Key Facts



Runs in Browser

No installation needed—works in every modern browser



Makes Pages Interactive

Transforms static content into dynamic experiences



Most Popular Language

Used by 98% of all websites



Analogy: If HTML is the skeleton and CSS is the appearance, JavaScript is the **nervous system** that makes everything respond!

JavaScript Example

Simple code that creates interaction

script.js

```
// Shows a popup message  
alert("Hello!");
```

Linking JS to HTML

```
<script src="script.js"></script>
```

What Happens?

1

Page Loads

Browser reads HTML and loads JavaScript file

2

JS Executes

`alert()` function runs immediately

3

Popup Appears

Browser displays message box with "Hello!"

Experiment

Try changing the message to "Welcome!" or your name. Add multiple alerts!

Frontend vs Backend

The two halves of full-stack development



Frontend

Client-Side

👁️ What Users See

Visual interface, layout, design, buttons, forms

</> Technologies

HTML, CSS, JavaScript, React, Vue

⚡ Responsibilities

User interface, interactivity, visual feedback

Analogy: Store **cashier**—what customers interact with directly



Backend

Server-Side

💡 Logic & Data

Business rules, calculations, data processing

</> Technologies

Node.js, Python, Java, Databases

🛡️ Responsibilities

Authentication, APIs, database management

Analogy: Store **stock room**—behind-the-scenes operations



Frontend



communicate



Backend



Full-Stack = Both!

| Getting Used to Cursor

Your AI-powered development environment



Cursor IDE

AI-First Code Editor

Cursor is a modern code editor built on VS Code with integrated AI assistance to help you write, understand, and debug code faster.

AI Features



Smart Code Completion

AI suggests code as you type



Chat with AI

Ask questions about your code



Error Detection

Real-time issue identification

Basic Workflow

1

Open Cursor

Launch the application

2

Create New Project

File → New Folder → Open

3

Create Files

index.html, style.css, script.js

4

Write & Save

Cmd/Ctrl + S to save changes

5

Run in Browser

Right-click → Open with Live Server



Pro Tip: Use **Ctrl+Space** for AI suggestions!

Practice Exercise

Apply what you've learned today

Your Tasks

1 Create HTML Page

Build a basic HTML structure with heading and paragraph

2 Add CSS Styles

Change colors, font sizes, and spacing

3 Add JavaScript

Create a simple alert or console message

4 Save Your Work

Organize files in a project folder

Success Criteria

 HTML page displays in browser

 CSS styling is applied correctly

 JavaScript runs without errors

 All files properly linked

Time Estimate

20-30 minutes. Don't worry about perfection—focus on understanding!



Need Help? Ask questions! Use Cursor's AI chat or raise your hand. 

Week 1, Day 1 Complete

Wrap-up

Great work today!

What We Learned

- ✓ What web applications are
- ✓ How the web works (browser ↔ server)
- ✓ HTML structure & tags
- ✓ CSS styling basics
- ✓ JavaScript interactivity
- ✓ Cursor IDE fundamentals

Optional Homework

Experiment with your practice project. Try changing colors, adding more HTML elements, or modifying the JavaScript alert message.

→ Tomorrow's Preview

We'll dive deeper into HTML—exploring more tags, attributes, and building more complex page structures!



You're on your way to becoming a Full-Stack Developer!

Keep practicing, stay curious, and don't hesitate to ask questions.

